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DEPARTMENT OF COMPUTER SCIENCE

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Topic : Types of activation functions and terminology of neural network

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Course Teacher

Introduction

An activation function is a mathematical function applied to the output of a neuron. It introduces non-linearity into a neural network, allowing the network to learn complex relationships between input and output.

Why Activation Functions Are Important

- They decide how strongly a neuron should be activated.
- They add non-linearity, which is essential for learning complex patterns.
- They influence gradient flow during backpropagation and therefore affect training speed.
- Different activation functions are suitable for different layers and tasks.

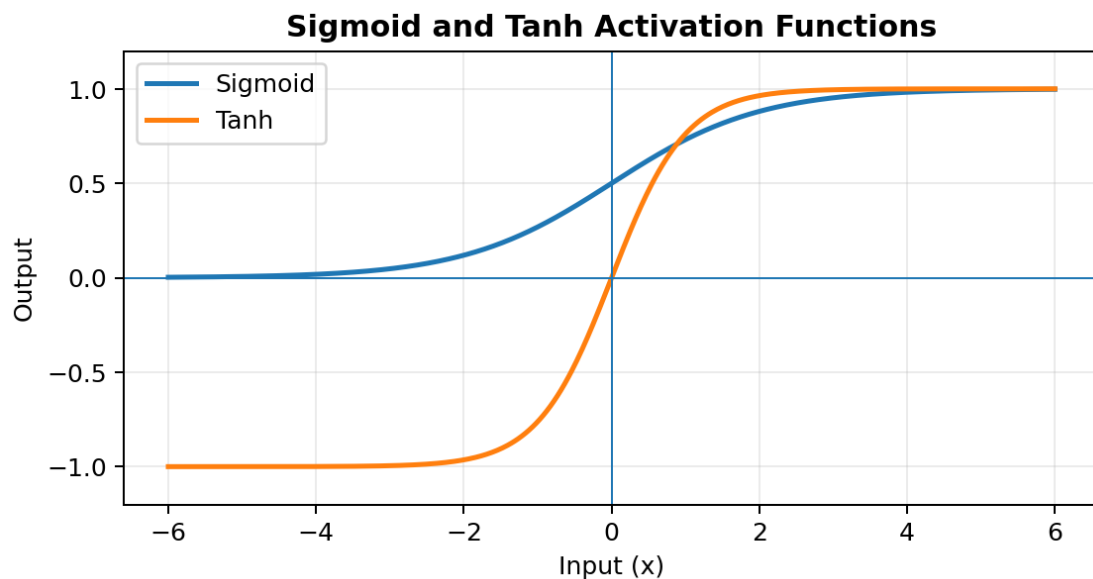
Main Types of Activation Functions

- Sigmoid: Produces values between 0 and 1; commonly used for binary output probabilities.
- Tanh: Produces values between -1 and 1 and is zero-centered.
- ReLU: Outputs zero for negative inputs and the input itself for positive inputs.
- Leaky ReLU: Keeps a small negative slope to reduce the dying-ReLU problem.
- Step: Produces a discrete output and is useful for simple threshold decisions.
- Linear: Returns the input directly and is often used for regression output layers.
- Softmax: Converts multiple class scores into probabilities whose sum is 1.

Basic Neuron Equation

A neuron first calculates a weighted sum and bias: $z = w_1x_1 + w_2x_2 + \dots + b$. The activation function then produces the neuron output: $a = f(z)$.

Sigmoid and Tanh Activation Functions



1. Sigmoid Function

Formula: $f(x) = 1 / (1 + e^{-x})$. The output lies between 0 and 1. It is easy to interpret as a probability, but very large positive or negative inputs can cause small gradients (the saturation problem).

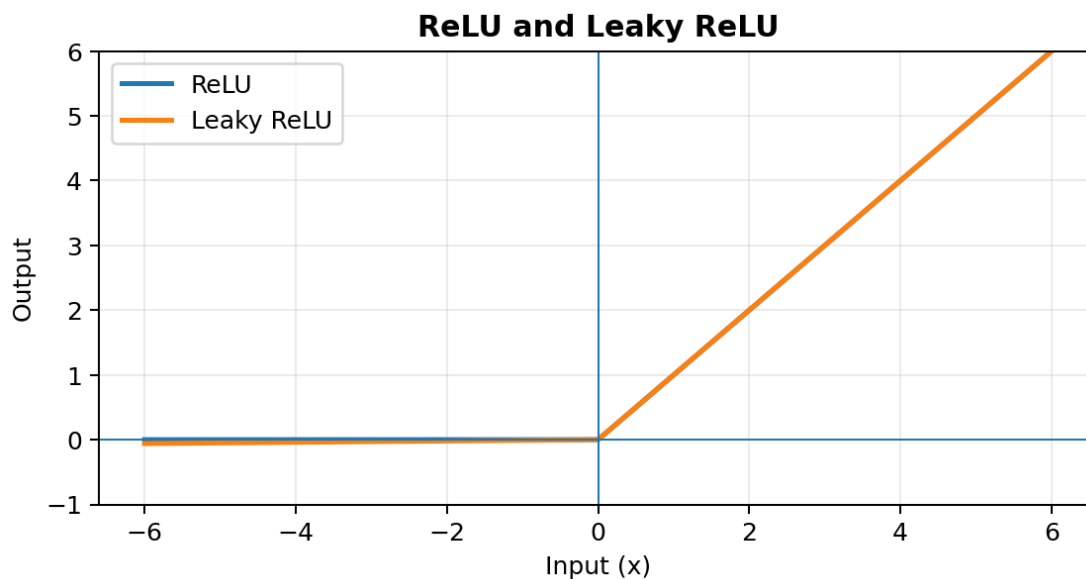
2. Tanh Function

Formula: $f(x) = \tanh(x)$. The output lies between -1 and 1 . It is zero-centered, which can make optimization more convenient than sigmoid in some hidden-layer settings.

Comparison

Sigmoid: range 0 to 1, useful for binary probability output. Tanh: range -1 to 1 , zero-centered. Both can suffer from vanishing gradients when saturated.

ReLU and Its Variants



1. ReLU (Rectified Linear Unit)

Formula: $f(x) = \max(0, x)$. ReLU is computationally simple and is widely used in hidden layers of deep neural networks. It gives zero for negative inputs and a positive linear response for positive inputs.

2. Leaky ReLU

Formula: $f(x) = x$ for $x \geq 0$ and αx for $x < 0$, where α is a small positive value. It allows a small gradient for negative inputs.

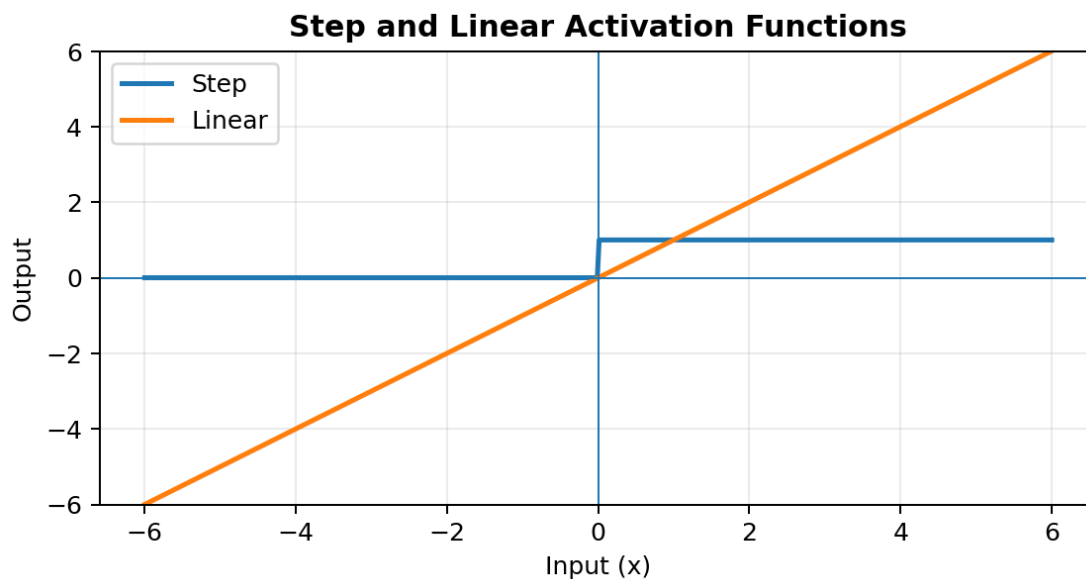
3. ELU and Related Variants

Other ReLU-family functions, such as ELU and Parametric ReLU, modify the negative region to improve gradient behavior or flexibility. The best choice depends on the architecture and training data.

Key Point

ReLU-family functions are especially popular in deep networks because they are simple and generally maintain useful gradients for positive inputs.

Step, Linear and Softmax Functions



1. Step Function

A step function gives a discrete output based on a threshold, for example 0 or 1. It is easy to understand but is not suitable for gradient-based training because its derivative is zero or undefined at the threshold.

2. Linear Function

Formula: $f(x) = x$. A linear activation is commonly used at the output of regression networks when the target can take a continuous range of values.

3. Softmax Function

Softmax converts a vector of class scores into probabilities. For class i , $\text{softmax}(z_i) = e^{z_i} / \sum_j e^{z_j}$. The probabilities are non-negative and sum to 1, making softmax common for multi-class classification.

Choosing an Activation Function

Hidden layers commonly use ReLU or related functions. Sigmoid is useful for binary probability outputs, softmax for multi-class outputs, and linear activation for many regression outputs. The choice should match the learning task.

Terminology of Neural Networks

Neuron / Node

A basic processing unit that receives inputs, applies weights and bias, and produces an output.

Input Layer

The first layer that receives the features or input values supplied to the network.

Hidden Layer

An intermediate layer that transforms information using weights, bias and activation functions.

Output Layer

The final layer that produces the network's prediction or decision.

Weight

A trainable value that controls the importance of an input connection.

Bias

A trainable offset added to the weighted sum, helping the neuron shift its activation.

Forward Propagation

The process of passing inputs through the network to calculate an output.

Loss Function

A function that measures the difference between predicted and actual values.

Backpropagation

A method for calculating gradients of the loss with respect to model parameters.

Gradient Descent

An optimization method that updates parameters in a direction that reduces the loss.

Learning Rate

A hyperparameter that controls the size of each parameter update.

Epoch

One complete pass through the training dataset during model training.